system that do not affect its gross macroscopic properties (such as its energy or pressure). The details are not essential so long as you realize that entropy is a fully quantitative quantum-mechanical concept that precisely measures the overall disorder of a physical system.

In 1970, Jacob Bekenstein, then a graduate student of John Wheeler's at Princeton, made an audacious suggestion. He put forward the remarkable idea that black holes might have entropy-and a huge amount of it. Bekenstein was motivated by the venerable and well-tested second law of thermodynamics, which declares that the entropy of a system always increases: Everything tends toward greater disorder. Even if you clean your cluttered desk, decreasing its entropy, the total entropy, including that of your body and the air in the room, actually increases. You see, to clean your desk you have to expend energy; you have to disrupt some of the orderly molecules of fat in your body to create this energy for your muscles, and as you clean, your body gives off heat, which jostles the surrounding air molecules into a higher state of agitation and disorder. When all of these effects are accounted for, they more than compensate for your desk's decrease in entropy, and thus the total entropy increases.

But what happens, Bekenstein in effect asked, if you clean your desk near the event horizon of a black hole and you set up a vacuum pump to suck all of the newly agitated air molecules from the room into the hidden depths of the black hole's interior? We can be even more extreme: What if the vacuum pumps all the air, and all the contents on the desk, and even the desk itself into the black hole, leaving you in a cold, airless, thoroughly ordered room? Since the entropy in your room has certainly decreased, Bekenstein reasoned that the only way to satisfy the second law of thermodynamics would be for the black hole to have entropy, and for this entropy to sufficiently increase as matter is pumped into it to offset the observed entropic decrease outside the black hole's exterior.

In fact, Bekenstein was able to draw on a famous result of Stephen Hawking's to strengthen his case. Hawking had shown that the area of the event horizon of a black hole—recall, this is the surface of no return that enshrouds every black hole—always increases in any physical interaction. Hawking demonstrated that if an asteroid falls into a black hole, or if some of the surface gas of a nearby star accretes